

7.3 Defining Tangent

Lesson Introduction

Lesson Introduction				
 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Target	I can use the ratios of sine and cosine to define the tangent ratio.
 Benchmark Coherence	Building On MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward N/A	
 Essential Questions	- How do sine and cosine define tangent? - How can I identify trigonometric ratios when given a reference angle? - How do I find missing sides of a right triangle?			
 Lesson Narrative	Through exploration, students will examine the relationship between sine and cosine to develop the tangent ratio. The tangent ratio is formally defined as ratio of $\frac{\text{sine}}{\text{cosine}}$ as well as $\frac{\text{opposite leg}}{\text{adjacent leg}}$. Students will find missing side lengths and identify the ratios of sine, cosine, and tangent.			
 Component Summary	7.3.1 Warm-Up (~5 min.)	Discuss notices and wonderings from a given image (<i>SE p. 382</i>)	7.3.3 Guided Instruction (10 min.)	Identify sine, cosine, and tangent when given a reference angle (<i>SE p. 384</i>)
	7.3.2 Exploration (15-20 min.)	Explore the relationship of sine and cosine to find tangent (<i>SE p. 382</i>)	7.3.4 Guided Practice (5-10 min.)	Solve for missing side lengths and identifying the ratios of sine, cosine, and tangent (<i>SE p. 386</i>)
	7.3.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 387</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Tangent <u>Additional Terms:</u> - Sine - Cosine - Reference angle - Adjacent leg - Opposite leg - Hypotenuse		(Optional) Calculator (Optional) Index Cards (Optional) Right triangles with no angles or sides labeled (Optional) Colored pencils	- Index cards with sine, cosine, and tangent written on them can provide support to kinesthetic learners when multiplying sine and cosine. - Consider providing copies of right triangles with no angles or sides labeled for students to use when no image is presented.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may multiply the numerators only when multiplying fractions. - Students may mislabel the opposite leg and the adjacent leg. - Students may forget to square the coefficient when squaring a radical expression.	Consider creating additional practice with right triangles that are not positioned with the right angle located at the end of a horizontal leg on the page.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk During 7.3.4 Guided Practice, allow students to share their different drawings and discuss why the drawings are correct. Use the opportunity to discuss how right triangles can be drawn differently but must always have a right angle, and how the hypotenuse is the longest side.	MA.K12.MTR.6.1: Reasonableness Throughout the lesson, discuss how the different representations of the ratio of sine, cosine, and tangent are correct. Facilitate a discussion with students about how simplifying radical expressions leads to equivalent ratios. Encourage students to check to make sure their answers are reasonable and correct.
 Differentiate Instruction	In 7.3.2 Exploration, allow students to color code the sides of the triangle in order to identify opposite, adjacent, and hypotenuse. Consider creating flashcards for students to use for a kinesthetic approach to multiplying and dividing fractions when discovering how sine multiplied by cosine will equal tangent. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

7.3.1 Warm-Up: Notice and Wonder

Component Overview: Using the image given, students will identify what they notice and wonder about the image.



7.3.1 Warm-Up: Notice and Wonder



1. What do you wonder?

Answers will vary.

Why is the faucet in the air?

How does the water get to the faucet?

2. What do you notice?

Answers will vary.

Faucet is in the air.

Water is pouring out.



Consider allowing a few students to share their responses. Do not spend too much time on this topic, but instead allow it to be used to explore growth mindset.

7.3.2 Exploration: Relating Sine, Cosine and Tangent

Component Overview: Students are exploring the relationship between sine and cosine to develop tangent.



7.3.2 Exploration: Relating Sine, Cosine and Tangent

- The table gives the sine and cosine values as decimals for different reference angle measures. Using a calculator and the values in the table, find the ratio of $\frac{\text{sine}}{\text{cosine}}$. Round to the nearest thousandth.

Angle Measure	Sine	Cosine	$\frac{\text{sine}}{\text{cosine}}$
10°	0.174	0.985	0.177
20°	0.342	0.940	0.364
30°	0.500	0.866	0.577
40°	0.643	0.766	0.839
50°	0.766	0.643	1.191
60°	0.866	0.500	1.732
70°	0.940	0.342	2.749
80°	0.985	0.174	5.661

- Use a calculator to find the tangent (tan) of each reference angle. Round to the nearest thousandth.

Angle Measure	Tan
10°	0.176
20°	0.364
30°	0.577
40°	0.839
50°	1.192
60°	1.732
70°	2.747
80°	5.671



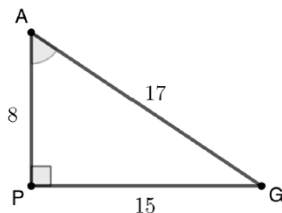
This Exploration is scaffolded to allow students to engage with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection.

7.3.2 Exploration: Relating Sine, Cosine and Tangent (continued)

3. Write down one notice statement and one wonder statement about the values of $\frac{\sin}{\cos}$ from the first table and the values of $\tan$ from the second table.

Notice	Wonder
<i>Answers will vary. I notice that the tangent value is approximately the same as the sine divided by cosine for the same angle measurements</i>	<i>Answers will vary. Why does the sine divided by the cosine result in the tangent?</i>

4. $\triangle APG$ is shown, where $m\angle APG = 90^\circ$, $AP = 8$, $PG = 15$, and $AG = 17$.



- a. Using $\angle PAG$ as the reference angle, complete the table.

Sine $\left(\frac{\text{opposite}}{\text{hypotenuse}}\right)$	Cosine $\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right)$	Tangent $\left(\frac{\text{opposite}}{\text{adjacent}}\right)$
$\frac{15}{17} \approx 0.882$	$\frac{8}{17} \approx 0.471$	$\frac{15}{8} = 1.875$

- b. Multiply the sine and the cosine ratios from the table.

$$\frac{15}{17} \cdot \frac{8}{17} = \frac{120}{289}$$



If students are finding that 10° is 0.648, then they have their calculators set in radian mode. Model for students how to change the calculator setting to degree.



Think-Pair-Share

Consider using this instructional routine for the Notice and Wonder. Allow students the opportunity to think about things they notice and wonder before sharing with a partner. A class discussion can ensure that students are noticing the relationship between the values of the ratio of sine and cosine and the values of tangent.



For a visual approach, allow students to color code the opposite leg, adjacent leg, and hypotenuse based on the reference angle. Using the same color coding, students can write the ratios for sine, cosine, and tangent.



Common Misconception

Students may get $\frac{120}{17}$ for the answer. Facilitate a discussion with the class about how to multiply fractions. If students are still struggling with fractions, consider allowing them to complete On-Ramp to Algebra 1: Multiplying Two Fractions.

7.3.2 Exploration: Relating Sine, Cosine and Tangent (continued)

- c. Rewrite $\text{sine} \cdot \text{cosine}$ using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. If possible, simplify.

$$\begin{aligned}\sin(\theta) \cdot \cos(\theta) &= \frac{\text{opposite}}{\text{hypotenuse}} \cdot \frac{\text{adjacent}}{\text{hypotenuse}} \\ &= \frac{\text{opposite}(\text{adjacent})}{\text{hypotenuse}^2}\end{aligned}$$

- d. Divide the sine ratio by the cosine ratio from the table.

$$\sin(\theta) \div \cos(\theta) = \frac{15}{17} \div \frac{8}{17} = \frac{15}{8}$$

- e. Rewrite $\frac{\text{sine}}{\text{cosine}}$ using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. The resulting ratio is a complex fraction that can be rewritten or simplified. Work with your partner to write the simplified form.

$$\frac{\sin(\theta)}{\cos(\theta)} = \frac{\frac{\text{opposite}}{\text{hypotenuse}}}{\frac{\text{adjacent}}{\text{hypotenuse}}} = \frac{\text{opposite}}{\text{hypotenuse}} \cdot \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{\text{opposite}}{\text{adjacent}}$$

- f. Describe how the ratios of sine and cosine can be used to find the tangent ratio.

Answers may vary. In order to find the tangent ratio of any right triangle, divide the ratio of sine by the ratio of cosine of the same reference angle.



For a kinesthetic approach, give students a set of 15 index cards with the following written on them:

- 5 - opposite
- 5 - adjacent
- 5 - hypotenuse

Students can use the cards to look at how to multiply and divide fractions instead of writing it out.



After question 4e, consider asking the following.

- What happened to the hypotenuse when dividing sine by cosine?
- If you are only given sine and cosine, can you find tangent? If so, how?
- If you are only given cosine and tangent, can you find the sine? If so, how?

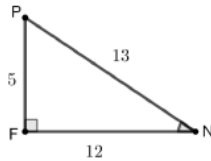
7.3.3 Guided Instruction: Determining Trigonometric Ratios

Component Overview: Students are finding and comparing the sine, cosine, and tangent from given reference angles.

7.3.3 Guided Instruction: Determining Trigonometric Ratios

The tangent ratio of a reference angle can be found using the ratio, $\frac{\text{sine}}{\text{cosine}}$.

1. $\triangle PFN$ is shown, where $m\angle PFN = 90^\circ$, $PF = 5$, $FN = 12$, and $PN = 13$.



- a. Using $\angle FNP$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{12}{13}$	$\frac{5}{13}$	$\frac{\left(\frac{5}{13}\right)}{\left(\frac{12}{13}\right)} = \frac{5}{13} \cdot \frac{13}{12} = \frac{5}{12}$

- b. Which method, $\frac{\text{sine}}{\text{cosine}}$ or $\frac{\text{opposite}}{\text{adjacent}}$, did you use to find the tangent ratio?

Answers may vary.

$$\frac{\text{sine}}{\text{cosine}} = \frac{\frac{5}{13}}{\frac{12}{13}} = \frac{5}{13} \cdot \frac{13}{12} = \frac{5}{12}$$

Or:

$$\frac{\text{opposite}}{\text{adjacent}} = \frac{5}{12}$$



If tangent is the ratio of sine to cosine, what sides of a right triangle can also be used to write a ratio for the tangent?



Common Misconception

Check to ensure students have correctly identified sine and cosine. If students have written the reciprocal of the ratios, then they may have mislabeled the sides. Consider having students color code the sides and/or draw an arrow from the reference angle to the side across from it in order to identify the opposite side.

7.3.3 Guided Instruction: Determining Trig Ratios (continued)

- c. Use the method for finding tangent that you did not use to check the ratio.

Answers may vary.

$$\frac{\text{sine}}{\text{cosine}} = \frac{\frac{5}{13}}{\frac{12}{13}} = \frac{5}{13} \cdot \frac{13}{12} = \frac{5}{12}$$

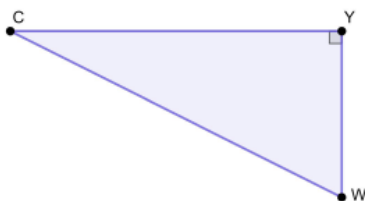
Or:

$$\frac{\text{opposite}}{\text{adjacent}} = \frac{5}{12}$$



Monitor student responses and consider having 2-3 students who used different methods share their responses.

2. $\triangle CYW$ is shown, where $CY = \sqrt{98}$ and $YW = \sqrt{28}$.



- a. Determine the length of $\overline{CW}$.

$$\sqrt{98}^2 + \sqrt{28}^2 = (CW)^2$$

$$CW = \sqrt{126} = 3\sqrt{14}$$

- b. Using $\angle WCY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{\sqrt{98}}{\sqrt{126}}$	$\frac{\sqrt{28}}{\sqrt{126}}$	$\frac{\sqrt{28}}{\sqrt{98}}$

- c. Using $\angle YWC$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{\sqrt{28}}{\sqrt{126}}$	$\frac{\sqrt{98}}{\sqrt{126}}$	$\frac{\sqrt{98}}{\sqrt{28}}$

7.3.3 Guided Instruction: Determining Trig Ratios (continued)

- d. What do you notice about the ratios for $\angle WCY$ and the ratios for $\angle YWC$?

Answers will vary.

The sine of one angle is the cosine of the other reference angle.

The tangent ratios are reciprocals. The numerator for the sine and cosine swap when swapping the reference angles.



Have students share with their partner or whole group what they notice about the ratios.

7.3.4 Guided Practice: Trigonometric Ratios

Component Overview: After using the Pythagorean Theorem to find missing sides of right triangles, students will write the ratios of sine, cosine, and tangent for a given reference angle.



7.3.4 Guided Practice: Trigonometric Ratios

1. $\triangle YNB$, $m\angle NBY = 90^\circ$, $BN = 6\sqrt{2}$, and $YN = 4\sqrt{17}$.

- a. Determine the length of $\overline{YB}$.

$$(6\sqrt{2})^2 + (YB)^2 = (4\sqrt{17})^2$$

$$YB = \sqrt{200} = 10\sqrt{2}$$

- b. Using $\angle BNY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{6\sqrt{2}}{4\sqrt{17}} = \frac{3\sqrt{2}}{2\sqrt{17}}$	$\frac{10\sqrt{2}}{4\sqrt{17}} = \frac{5\sqrt{2}}{2\sqrt{17}}$	$\frac{10\sqrt{2}}{6\sqrt{2}} = \frac{5}{3}$

2. $\triangle GMK$, $m\angle GKM = 90^\circ$, $GM = 3\sqrt{10}$, and $GK = 3\sqrt{3}$.

- a. Determine the length of $\overline{KM}$.

$$(3\sqrt{3})^2 + (KM)^2 = (3\sqrt{10})^2$$

$$KM = \sqrt{63} = 3\sqrt{7}$$

- b. Using $\angle KGM$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent
$\frac{3\sqrt{3}}{3\sqrt{10}} = \frac{\sqrt{3}}{\sqrt{10}}$	$\frac{3\sqrt{7}}{3\sqrt{10}} = \frac{\sqrt{7}}{\sqrt{10}}$	$\frac{3\sqrt{7}}{3\sqrt{3}} = \frac{\sqrt{7}}{\sqrt{3}}$



Provide students with copies of right triangles with no sides or angles labeled. Task students with appropriately labeling the triangle prior to answering the questions.



Common Misconception

If students find the missing side to be $\sqrt{3}$, then students did not square 3 when squaring $3\sqrt{10}$. Facilitate a discussion about how to square radicals when there is a coefficient.

7.3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

Answers will vary.



The information provided here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.